Code No: 117FN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, June/July - 2024
MOBILE COMPUTING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the limitations of mobile computing model?	[2]
b)	What are the differences between traffic and control channels in GSM?	[3]
c)	What are the different entities in mobile IP?	[2]
d)	What is meant by hidden terminal problem?	[3]
e)	What is slow start in TCP?	[2]
f)	Why traditional TCP is not suitable in mobile computing?	[3]
g)	Explain the significance of data dissemination for mobile hosts.	[2]
h)	Mention the types of data synchronization in mobile computing.	[3]
i)	What are the characteristics of PalmOS?	[2]
j)	What are the characteristics of Mobile Adhoc Networks?	[3]

PART – B

(50 Marks)

2. Describe about the protocol architecture of DECT. [10]

OR

3.a) Explain about the key generation algorithm in GSM security model.
b) What are the applications of Mobile Computing? [5+5]

4. What is the basic purpose of DHCP? Name the entities of DHCP. How can DHCP be used for mobility support. [10]

OR

5.a) Explain about the Generic Routing Encapsulation.
b) Explain exposed and hidden terminals and near and far terminal problems of wireless transmission. [5+5]

6. Illustrate how I-TCP overcomes the problems associated with TCP in mobile environment. [10]

OR

7.a) Explain about the Data Recovery Process in mobile databases.
b) Explain about Quality of Service issues in mobile computing. [5+5]

8.a) What is Asymmetric communication? How does it work in mobile computing?
b) Explain about the push based data delivery mechanism. [5+5]

OR

9.a) Explain about the pull based data delivery mechanism.
b) What is the motivation for selective turning by mobile hosts? [5+5]

10. Explain the protocol architecture of Bluetooth. [10]

OR

11. Explain how to find the route from source to destination in DSDV protocol and also illustrate with an example. [10]

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